

DEPARTMENT OF MATHEMATICS

MATH 1210 Techniques of Classical and Linear Algebra

May – June 2023

Instructor:	Romain Branchereau, 452 Machray Hall, Tel: 204-474-6686 , e-mail: Romain.Branchereau@umanitoba.ca
Lecture:	Armes 205, Monday, Tuesday, Wednesday, Thursday 9:00 – 10:15
Office hours:	Thursday 11:00–12:00, Friday 9:30-10:30, or by appointment
Labs:	B01, Armes 115, Monday and Wednesday 10:45 – 11:35 B02, Armes 111, Monday and Wednesday 10:45 – 11:35

Pre-Requisites: a minimum grade of 60% in Pre-calculus Mathematics 40S or the former Mathematics 40S (300), or a grade of "C" or better in the MSKL 100 offered by Extended Education.

Textbook: *Notes for Course MATH 1210: Techniques of Classical and Linear Algebra by Donald Trim (2012)*

Course Outline:

1. **Mathematical Induction:** In this unit proof by mathematical induction, and use of sigma notation to represent sums symbolically will be discussed (Chapter 1).
2. **Complex Numbers and Polynomial Equations:** This unit covers complex numbers in Cartesian, polar, and exponential forms, addition, subtraction, multiplication, division, and roots of complex numbers (Chapter 2), properties of polynomial equations, and use the rational root theorem, Descartes' rules of signs, and a bounds theorem for finding roots of polynomial equations (Chapter 3).
3. **Linear Algebra:** This unit covers a variety of topics in linear algebra, such as matrices and matrix arithmetic (Chapter 4), vectors and their application to equations for lines and planes in three-space (Chapter 5), solving systems of linear equations by Gaussian and Gauss-Jordan elimination (Chapter 6), determinants, Cramer's Rule, linearly independent and dependent vectors (Chapter 7), inverses of matrices, using inverse matrices to solve linear system (Chapter 8), linear transformations, matrices for linear transformations and eigenvalues and eigenvectors (Chapter 9).

Tutorials: There are two lab sections. You must be registered and attend in one of them (B01 or B02). They both take place on Monday and Wednesday from 10:45AM to 11:35AM. The tutorials start on the first week, the first sessions are on Monday, May 8.

Student Evaluation:

- **Term tests:** There will be three 50 minutes in-person Crowdmark examination. The average of your best 2 out of 3 will be worth 45% of the final grade (22.5% each). They will be conducted during lab time **10:45 to 11:35** in **Armes 204** on

Wednesday, May 17 and Friday, May 26 and Monday, June 5 .

- **Final exam:** A two hour in-person Crowdmark final examination, to be scheduled by the Registrar's Office, will count for the remaining 55% of the final grade.
- **Bonus for participation:** During the tutorials you will be given time to work on some exercises and the possibility to present your solution to the class. You will then get a bonus point if your solution is complete and serious, even if contains mistakes. The points that you collect during the term will give

you the following bonus at the end of the term: the first point gives you 1% on the final percentage that will determine the letter grade, the remaining points will give you 0.25% on the final percentage. The maximal bonus that you can obtain is 3%. For example if you participate three times during the term, you will get a bonus of 1.5%.

Grading Table: The following is the grading table which may be adjusted downwards:

Letter Grade	Minimum percentage to guarantee	Final Grade Point
A+	93	4.5
A	83	4.0
B+	77	3.5
B	71	3.0
C+	65	2.5
C	60	2.0
D	50	1.0

Notes:

1. The term tests will take place during the tutorial time in Armes 204 and will last 50 minutes. The schedule for the tests is as follows:

Test	Date	Time	Place	Topics (may vary slightly)
1	Wed, May 17	10:45-11:35	Armes 204	Chapters 1 and 2
2	Fri, May 26	10:45-11:35	Armes 204	Chapters 1 to 5
3	Mon, June 5	10:45-11:35	Armes 204	Chapters 1 to 7

Marked term tests will be given back to the students, with mark and comments, in approximately 10 days after submission through Crowdmark.

2. If you miss a term test, you will be assigned a mark of "zero". There will be no make-up exam.
3. Appeals on the tests results have to be made within 7 days after the results.
4. Attendance is an important part of taking this course and students are expected to attend all classes and are responsible for all material presented in class even if they are not present. In case of missing a lecture for acceptable reason students need to contact their instructor for a copy of lecture notes.
5. Calculators or other computing devices are not allowed for the term tests or the final exam.
6. Student should not participate in personal direct electronic messaging / posting activities (e-mail, texting, video or voice chat, wikis, blogs, social networking (e.g. Facebook) online and offline "gaming" during scheduled class time.
7. Learn Efficiently, Learn to Take Notes: During the pandemic, many lectures were delivered asynchronously so students had a chance to review lecture videos when they did not catch something during the lecture. Lectures are delivered in-person this term and students will not have the luxury of rewinding a lecture. Therefore, you may want to review some note-taking tips offered by the Academic Learning Centre which can help you learn efficiently. (A copy of tips is provided in the course web page in UM Learn)
8. Commitment to Study: In order to be successful in this course, you should aim to commit at least three hours of studying for every hour of lecture. Make sure you keep up with studying on a consistent basis.

9. The instructors hold copyright over the course materials, presentations and lectures which form part of this course. No audio or video recording of lectures or presentations is allowed in any format, openly or surreptitiously, in whole or in part without permission. Course materials (both paper and digital) are for the participant's private study and research.
10. If you are a student with a disability, please contact SAS for academic accommodation supports and services such as note-taking, interpreting, assistive technology and exam accommodations. Students who have, or think they may have, a disability (e.g. mental illness, learning, medical, hearing, injury-related, visual) are invited to contact SAS to arrange a confidential consultation.

Student Accessibility Services <http://umanitoba.ca/student-supports/accessibility>
 520 University Centre
 204 474 7423
student_accessibility@umanitoba.ca

11. Should your instructor be ill, he/she will work with the Mathematics Department to find a suitable replacement to teach the in-person class as soon as possible. In the event a suitable instructor cannot be found, your instructor may give the lecture synchronously through Zoom or WebEx for up to 5 days. An email will be sent to students' UM email and a posting will be made on the course web page on UM Learn the night before class if this is the case.
12. Voluntary Withdrawal Deadline is Thursday June 1, 2023.

Tentative schedule and important dates

Week	Date	Evaluation	Material covered in class
1	May 8 – May 11		1.1, 1.2, 2.1, 2.2
2	May 15 – May 18	Test 1: Wednesday, May 17	2.2, 3.1, 3.2, 4.1
3	May 23 – May 26	Test 2: Friday May 26	4.2, 5.1, 5.2, 6.1, 6.2
4	May 29 – June 1	VW: Thursday June 1	6.3, 6.4, 7.1, 7.2
5	June 5 – June 8	Test 3: Monday, June 5	7.3, 8.1, 8.2, 8.3, 8.4, 8.5
6	June 12 – June 15		9.1, 9.2, 9.3, 9.4

Web Page for Archived Materials

Dr. R.S.D. Thomas has archived materials for this course over a number of years. This archive contains many assignments, midterm exams and final exams for the last 15 years. It can be found at <http://home.cc.umanitoba.ca/~thomas/Courses>

Useful Links

UM Learn : <https://centre.cc.umanitoba.ca/technology/umlearn/>.

Crowdmark : <https://crowdmark.com/help/categories/support-for-students/>.

Academic Integrity

The Department of Mathematics, The Faculty of Science and The University of Manitoba regard acts of academic dishonesty in quizzes, tests, examinations, laboratory reports or assignments as serious offences and may assess a variety of penalties depending on the nature of the offence.

Acts of academic dishonesty include, but are not limited to bringing unauthorized materials into a test or exam, copying from another individual, using answers provided by tutors, plagiarism, and examination personation.

Note: Pagers, PDAs, MP3 units or electronic translators are explicitly listed as unauthorized materials, and must not be present during tests or examinations.

Penalties for violation include being assigned a grade of zero on a test or assignment, being assigned a grade of “F” in a course, compulsory withdrawal from a course or program, suspension from a course/program/faculty, or even expulsion from the University. Further information about suggested minimum penalties assessed by the Faculty of Science can be found here: <https://sci.umanitoba.ca/statement-on-academic-dishonesty/>.

All students are advised to familiarize themselves with the **Student Discipline Bylaw**, which is printed in its entirety in the Student Guide; also available online or through the Office of the University Secretary.

Notice Regarding Personal Information And Crowdmark

Your personal information is being collected under the authority of The University of Manitoba Act. It will be used for the purposes of grading papers and providing feedback to students. Personal information will not be used or disclosed for other purposes, unless permitted by The Freedom of Information and Protection of Privacy Act (FIPPA). The University of Manitoba has taken steps to ensure that its agreements with Crowdmark, Inc. for services provided by the Crowdmark application are in compliance with FIPPA. Please be aware that information held by Crowdmark Inc. may be transmitted to and stored on servers outside of the University of Manitoba, or Canada. The University of Manitoba cannot and does not guarantee protection against the possible disclosure of your data including, without limitation, against possible secret disclosures of data to a foreign authority in accordance with the laws of another jurisdiction. If you have any questions about the collection of personal information, contact the Access and Privacy Office (tel. 204-474-9462), The University of Manitoba, 233 Elizabeth Dafoe Library, Winnipeg, Manitoba, Canada, R3T 2N2.